BITS PILANI WILP — M.TECH (AI & ML)

Course: Data Mining and Machine Learning (DMML)

RecoMart Recommendation Pipeline

Phase 1: Problem Formulation Report

1. Problem Formulation

1.1 Business Problem Definition:
In modern digital platforms, information overload degrades user experiences, leading to high bounce rates and lost engagement opportunities. The objective of the RecoMart Recommendation System is to maximize user engagement and conversion rates by presenting personalized, highly relevant catalog suggestions. This project implements a Collaborative Filtering pipeline using an explicit matrix factorization approach (SVD) to model historical user-item interactions and predict ratings on unseen inventory. By serving these recommendations, the platform directly optimizes click-through rates (CTR) and customer retention.

1.2 Key Data Sources and Attributes:

The architecture ingests data from two distinct channels to simulate an enterprise ecosystem: a static historical interaction dataset (MovieLens 100K) and an external live product catalog API.

Entity / Dataset	Key Attributes & Descriptions
Users (u.user / customers.csv)	• customer_id (Unique Key) • age (Integer used for cohort binning) • gender (Categorical) • occupation (Categorical professional status)
Items (u.item / articles.csv)	• article_id (Unique Key) • movie_title (String descriptive metadata) • release_date (Temporal feature) • genre_0 to genre_18 (19 Binary genre flags)
Interactions (u.data / transactions.csv)	• customer_id (User foreign key) • article_id (Item foreign key) • rating (Explicit feedback score on [1, 5] scale) • timestamp (Epoch time)
External Catalog API (products_api.csv)	• id (API source primary key) • title, price, category, rating_rate (Auxiliary dynamic properties)

1.3 Expected Pipeline Outputs:

The modular machine learning pipeline isolates execution steps to deliver clear, versioned output artifacts:

- **Clean Datasets for EDA:** De-duplicated tables free of age/rating outliers, median-imputed values, and standardized visual summary charts showing user age densities, movie popularity counts, and rating sparsity matrices.
- **Engineered Features:** Aggregated user features (such as user interaction counts and student status indicators) coupled with timestamp components parsed into offline Feature Store structures.
- **Deployable Model & Inference Assets:** A serialized matrix-factorization SVD model binary (recommendation_model.pkl) and a high-performance script delivering targeted Top-10 output tables (recommendations.csv) for end-consumers.

1.4 Pipeline Evaluation Metrics:

Performance is tracked across two primary paradigms:

- **Offline Prediction Quality (RMSE):** Evaluates how closely predicted rating scores map to real user values.
- **Retrieval Ranking Metrics (Precision@K and Recall@K, where K=10):** Evaluates the system's accuracy in placing highly rated, relevant items within the user's top recommended slots.